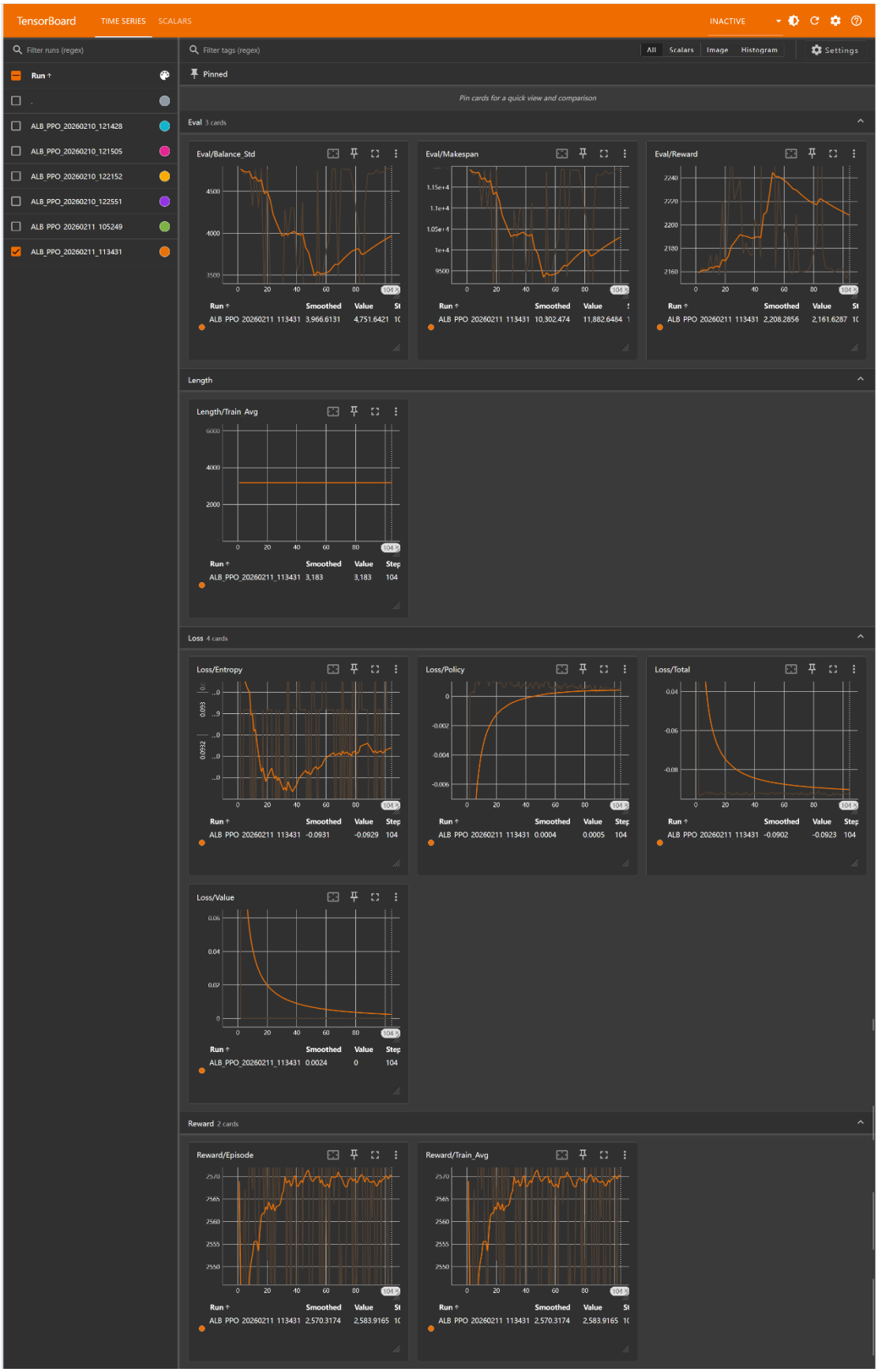
V4.1

533 机

```
seed = 42
n_j = 3299 # Tasks (Dataset 3000.xlsx)
n_w = 50 # Workers (Increased for large dataset)
# Model
hidden_dim = 256 # Reduced from 128 for speed/memory on 3000 tasks
num_gat_layers = 3
num_heads = 4
dropout = 0.15

# Feature Dimensions (Estimated, will be updated by data_loader)
task_feat_dim = 17
worker_feat_dim = 12
station_feat_dim = 8

# Training
lr = 1e-4
gamma = 0.99
k_epochs = 1 # [Optimization] Reduced for speed with large batch
eps_clip = 0.2
max_updates = 1000
batch_size = 2 # [Fix OOM] Small batch size for update
```



记录：在 60 步结果比较好

V4.3

869 机

```
# Environment
seed = 42
n_j = 3299 # Tasks (Dataset 3000.xlsx)
n_m = 5    # Stations (Estimated, will be loaded from data)
n_w = 50 # Workers (Increased for large dataset)
data_file_path = "3000.csv"

# Model
hidden_dim = 128 # Reduced from 128 for speed/memory on 3000 tasks
num_gat_layers = 3
num_heads = 4
dropout = 0.1

# Feature Dimensions (Estimated, will be updated by data_loader)
task_feat_dim = 17
worker_feat_dim = 12
station_feat_dim = 8

# Training
lr = 1e-4
gamma = 0.99
k_epochs = 1 # [Optimization] Reduced for speed with large batch
eps_clip = 0.2
max_updates = 3000
```

batch_size = 1 # [Fix OOM] Small batch size for update

V4.4

597 机

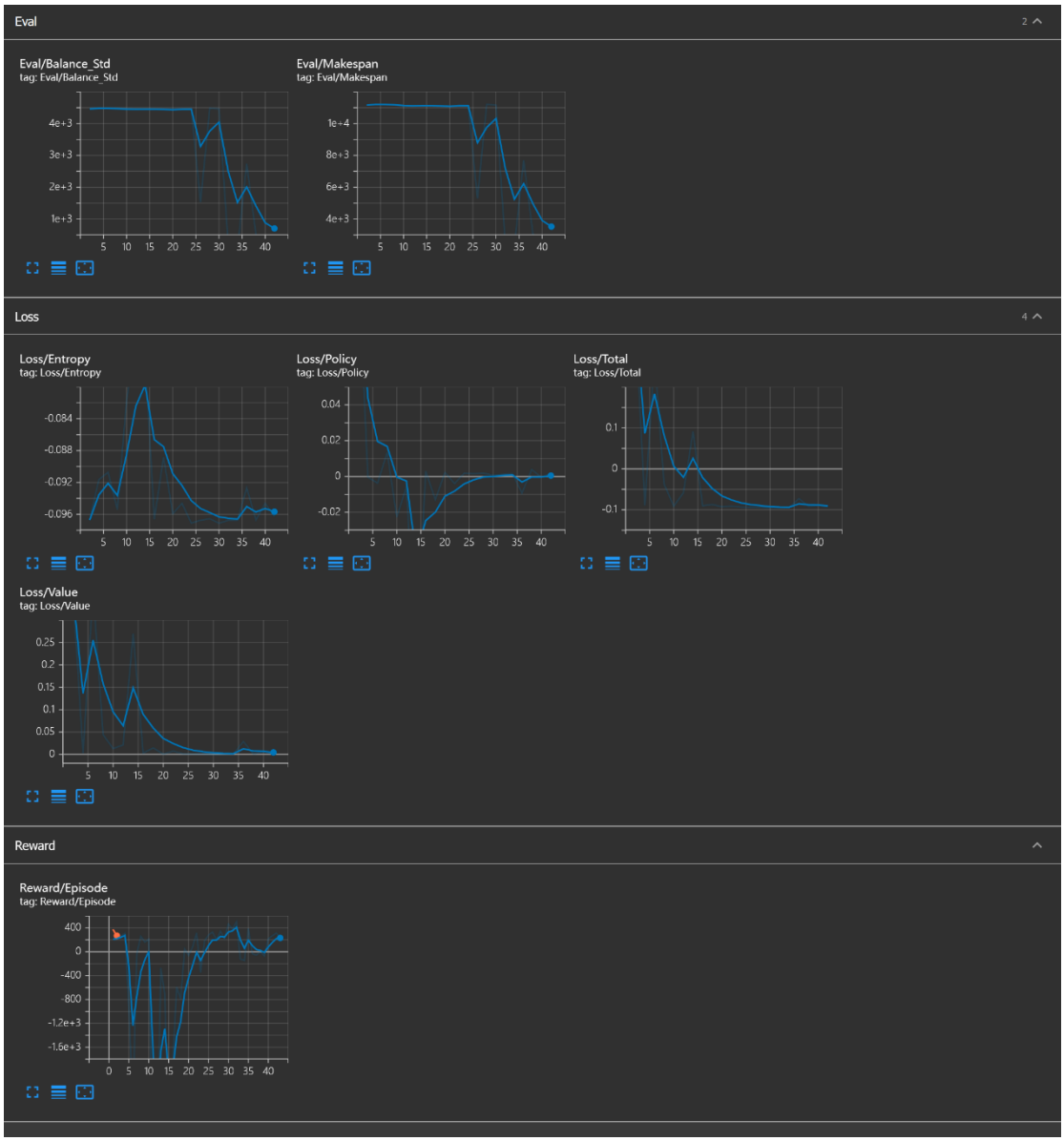
```
seed = 42                       # 随机种子：确保训练和验证环境的一致性，防止过拟合
n_j = 3299                      # 任务总数：根据 3000.csv 数据集确定
n_m = 5                         # 站位数：假设流水线有 5 个工位
n_w = 50                        # 工人总数：为了适应大规模任务集，设置了 50 人
data_file_path = "3000.csv"     # 数据集路径

# ------------------
# 模型架构超参数 (Model Hyperparameters)
# ------------------
hidden_dim = 128                # 隐藏层维度：控制模型的表达能力。过大易OOM，过小欠拟合。
num_gat_layers = 5              # GAT层数：控制图神经网络的消息传递深度。3层通常能捕获较好的局部依赖。
num_heads = 4                   # 注意力头数：多头注意力机制，提升对不同关系的捕捉能力。
dropout = 0.1                   # Dropout比率：防止过拟合，增强模型鲁棒性。

# ------------------
# 特征维度 (Feature Dimensions) - 运行时会根据 data_loader 自动校准
# ------------------
task_feat_dim = 17              # 任务特征维度：包含 Duration, SkillType, Status, WaitTime, Demand 等
worker_feat_dim = 12            # 工人特征维度：包含 Efficiency, SkillVector, IsFree 等
station_feat_dim = 8            # 站位特征维度：包含 Load, NumTasks 等

# ------------------
# PPO 训练超参数 (Training Hyperparameters)
# ------------------
lr = 1e-4                       # 学习率：决定梯度下降的步长。3e-4 是 PPO 的经典默认值。
gamma = 0.99                    # 折扣因子：决定智能体对未来奖励的重视程度。0.99 表示远视。
k_epochs = 1                    # PPO 更新次数：每批数据在这个策略上更新几次。设为 1 是为了在大规模任务下加速。
eps_clip = 0.2                  # PPO 截断范围：限制策略更新幅度，保证训练稳定性。
max_updates = 3300              # 最大训练轮次：Episodes 总数。
batch_size = 2                  # PPO Batch Size：由于 Graph 数据较大，设为 1 避免显存溢出 (OOM)。

# ------------------
# 日志与监控 (Logging)
# ------------------
log_dir = "tf-logs"                # TensorBoard 日志目录：存储 loss, reward 等训练曲线。
```

Eval
Eval/Balance_Std
tag: Eval/Balance_Std
Eval/Makespan
tag: Eval/Makespan
Loss
Loss/Entropy
tag: Loss/Entropy
Loss/Policy
tag: Loss/Policy
Loss/Total
tag: Loss/Total
Loss/Value
tag: Loss/Value
Reward
Reward/Episode
tag: Reward/Episode

V4.5

869 机

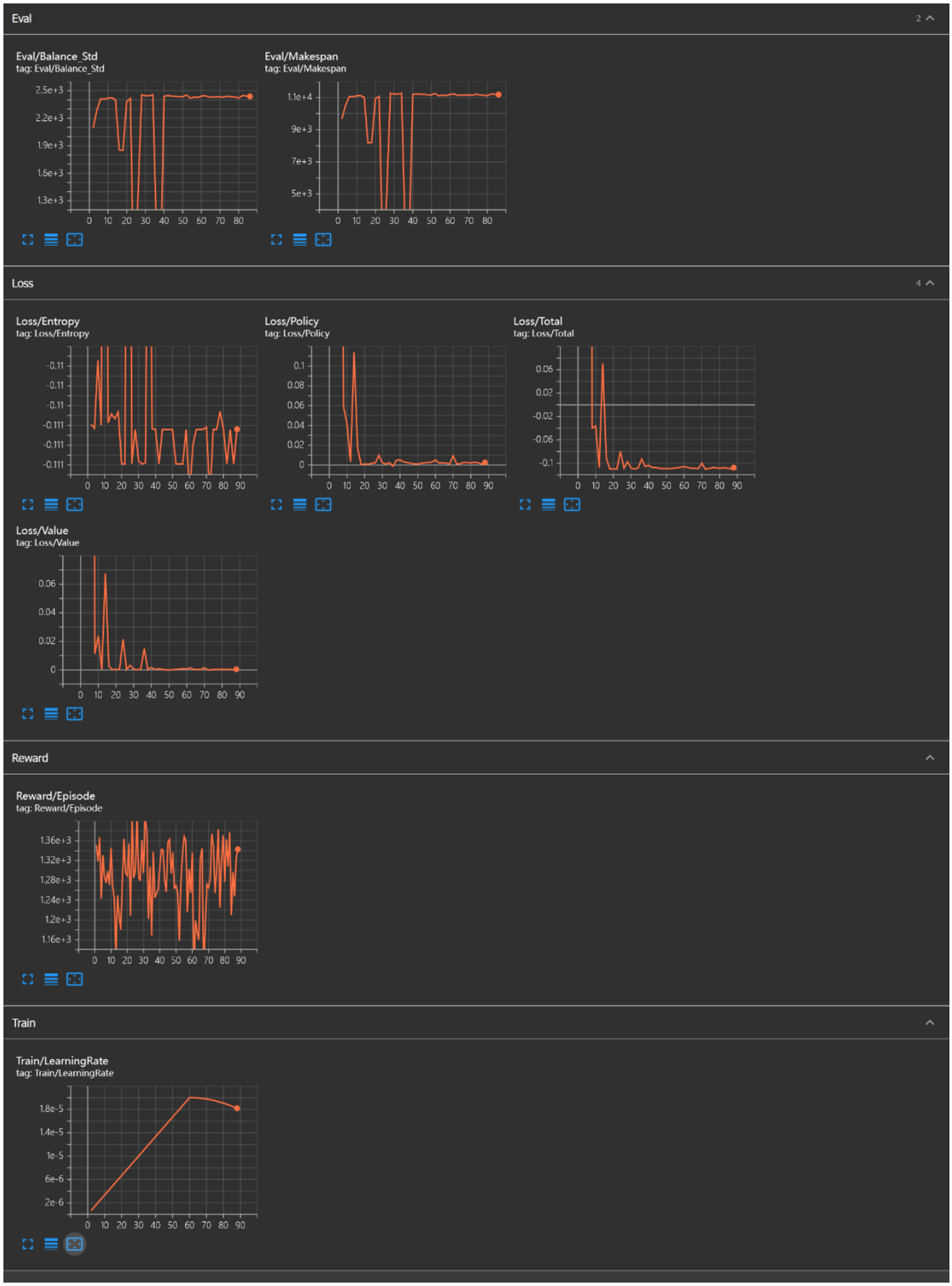
```python
# ------------------
seed = 42                       # 随机种子：确保训练和验证环境的一致性，防止过拟合
n_j = 3299                      # 任务总数：根据 3000.csv 数据集确定
n_m = 5                         # 站位数：假设流水线有 5 个工位
n_w = 60                        # 工人总数：为了适应大规模任务集，设置了 50 人
data_file_path = "3000.csv"     # 数据集路径

# ------------------
# 模型架构超参数 (Model Hyperparameters)
# ------------------
hidden_dim = 128                # 隐藏层维度：控制模型的表达能力。过大易OOM，过小欠拟合。
num_gat_layers = 5              # GAT层数：控制图神经网络的消息传递深度。3层通常能捕获较好的局部依赖。
num_heads = 4                   # 注意力头数：多头注意力机制，提升对不同关系的捕捉能力。
dropout = 0.12                   # Dropout比率：防止过拟合，增强模型鲁棒性。

# ------------------
# 特征维度 (Feature Dimensions) - 运行时会根据 data_loader 自动校准
# ------------------
task_feat_dim = 17              # 任务特征维度：包含 Duration, SkillType, Status, WaitTime, Demand 等
worker_feat_dim = 12            # 工人特征维度：包含 Efficiency, SkillVector, IsFree 等
station_feat_dim = 8            # 站位特征维度：包含 Load, NumTasks 等

# ------------------
# PPO 训练超参数 (Training Hyperparameters)
# ------------------
lr = 3e-5                       # 学习率：决定梯度下降的步长。3e-4 是 PPO 的经典默认值。
gamma = 0.99                    # 折扣因子：决定智能体对未来奖励的重视程度。0.99 表示远视。
k_epochs = 1                    # PPO 更新次数：每批数据在这个策略上更新几次。设为 1 是为了在大规模任务下加速。
eps_clip = 0.2                  # PPO 截断范围：限制策略更新幅度，保证训练稳定性。
max_updates = 3300              # 最大训练轮次：Episodes 总数。
batch_size = 2                  # PPO Batch Size：由于 Graph 数据较大，设为 1 避免显存溢出 (OOM)。
```


Eval
Eval/Balance_Std
tag: Eval/Balance_Std
Eval/Makespan
tag: Eval/Makespan
Loss
Loss/Entropy
tag: Loss/Entropy
Loss/Policy
tag: Loss/Policy
Loss/Total
tag: Loss/Total
Loss/Value
tag: Loss/Value
Reward
Reward/Episode
tag: Reward/Episode

在 40 步左右达到最好

V4.6

```
# ------------------
n_j = 3300                      # 任务(工序)数量估计 (Graph Nodes)
n_m = 20                        # 站位数量 (Stations)
n_w = 60                        # 工人数量 (Workers)
                                # 注意：实际任务数由 data_loader 动态加载，此处仅作参考或 Embedding 初始化上界

# ------------------
# 模型超参数 (Model Hyperparameters)
# ------------------
hidden_dim = 128                # 隐藏层维度 (Embedding Size)
num_gat_layers = 5              # GAT 层数 (Message Passing Depth)
num_heads = 4                   # 多头注意力头数 (Attention Heads)
dropout = 0.12                  # Dropout 比率 (防止过拟合)

# 输入特征维度 (根据 environment.py 中的 _get_observation 定义)
task_feat_dim = 17              # Task Node Input Features [Duration, Status(4), Type(10), Wait(1), Demand(1)]
worker_feat_dim = 12            # Worker Node Input Features [Efficiency(1), Skills(10), IsFree(1)]
station_feat_dim = 8            # Station Node Input Features [Load(1), NumTasks(1), Padding(6)]

# ------------------
# PPO 训练超参数 (PPO Training)
# ------------------
lr = 2e-5                       # 初始学习率 (Learning Rate)
gamma = 0.99                    # 折扣因子 (Discount Factor)
k_epochs = 1                    # 每次更新循环次数 (PPO Update Epochs)
eps_clip = 0.2                  # PPO Clip阈值 (e.g. 0.1 ~ 0.2)
batch_size = 2                  # PPO update batch size (小批量更新)

# [Training Control Parameters 2026-02-12]
max_episodes = 200             # 最大训练 Episode 数
update_every_episodes = 2       # 多少个 Episode 收集一次数据进行 PPO 更新
eval_freq = 2                  # 多少个 Episode 进行一次评估

# [Learning Rate Schedule]
lr_warmup_steps = 30           # 学习率预热步数 (Linear Warmup)
min_lr = 1e-6                   # 最小学习率 (Cosine Annealing 下界)
```



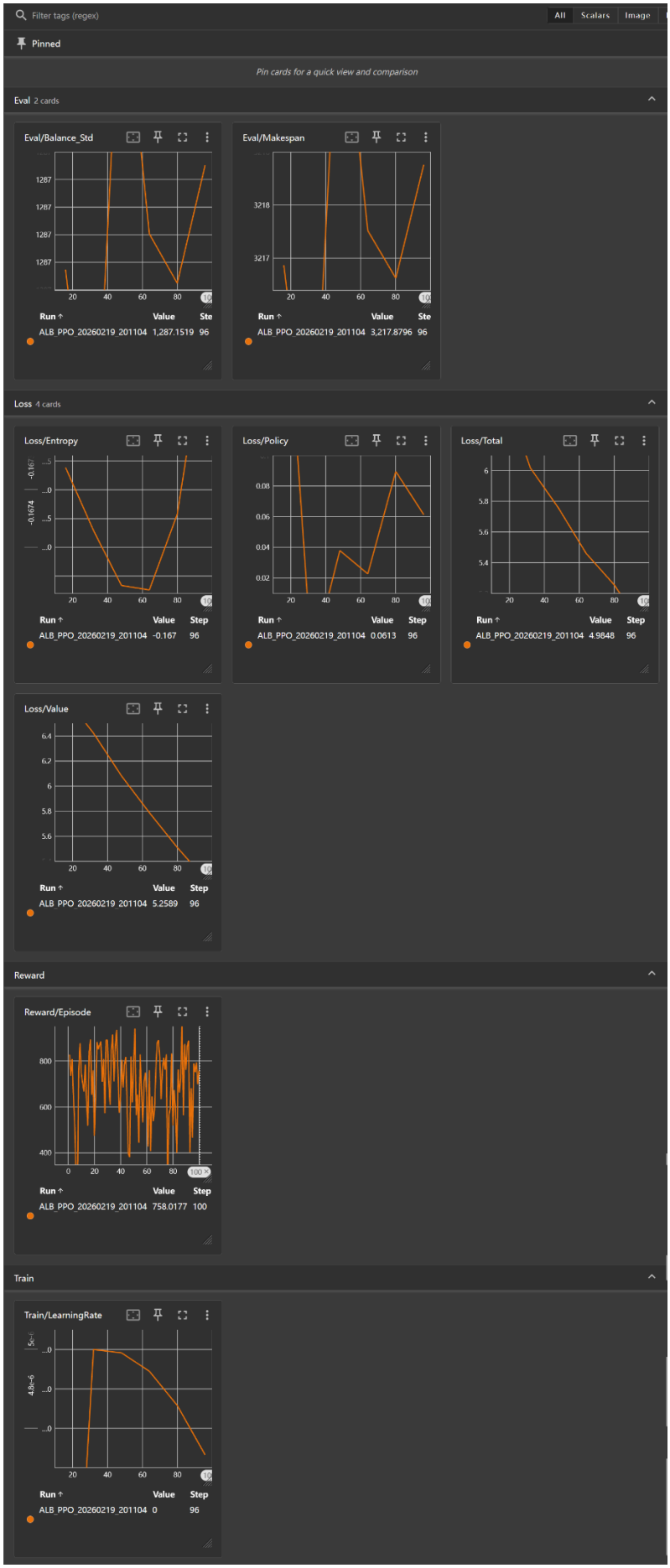
V4.7/v4.8

class configs:

```
# ------------------
# 路径配置 (Paths)
# ------------------
# 默认数据路径 (如果没有通过命令行参数指定)
data_file_path = "100.csv"

# ------------------
# 环境与图相关 (Environment & Graph)
# ------------------
n_j = 715                    # 任务(工序)数量估计 (Graph Nodes)
n_m = 5                      # 站位数量 (Stations)
n_w = 30                     # 工人数量 (Workers)
# ------------------
# 模型超参数 (Model Hyperparameters)
# ------------------
hidden_dim = 128             # 隐藏层维度 (Embedding Size)
num_gat_layers = 5           # GAT 层数 (Message Passing Depth)
num_heads = 4                # 多头注意力头数 (Attention Heads)
dropout = 0.12               # Dropout 比率 (防止过拟合)

# 输入特征维度 (根据 environment.py 中的 _get_observation 定义)
task_feat_dim = 17           # Task Node Input Features [Duration, Status(4), Type(10), Wait(1), Demand(1)]
worker_feat_dim = 12         # Worker Node Input Features [Efficiency(1), Skills(10), IsFree(1)]
station_feat_dim = 8         # Station Node Input Features [Load(1), NumTasks(1), Padding(6)]
```

```
# ------------------
# PPO 训练超参数 (PPO Training)
# ------------------
lr = 5e-6                          # 初始学习率 (Learning Rate)
gamma = 0.99                       # 折扣因子 (Discount Factor)
k_epochs = 2                       # [Tuning] 增加更新力度 (1 -> 4)
eps_clip = 0.2                     # PPO Clip 阈值 (e.g. 0.1 ~ 0.2)
batch_size = 16                    # PPO update batch size (物理显存限制)
accumulation_steps = 64            # [GA] 梯度累积步数 (逻辑 Batch Size = 2 * 32 = 64)
entropy_coef = 0.02                # [Tuning] 熵正则化系数 (强迫探索)

# [Training Control Parameters 2026-02-12]
max_episodes = 100                 # 最大训练 Episode (Reduced from 2000 -> 150)
update_every_episodes =16          # 多少个 Episode 收集一次数据进行 PPO 更新
eval_freq = 16                     # 多少个 Episode 进行一次评估
# 用户目标: Ep 20 达到峰值并开始衰减
lr_warmup_episodes = 5             # 预热 20 episodes
lr_hold_episodes = 4               # 峰值保持 5 个 Episode
min_lr = 1e-7                      # 最小学习率

# [Reward Function Tuning]
reward_baseline = 1000.0           # 基准 Makespan (用于计算正向奖励)
reward_scale = 2                   # 奖励缩放系数 (Reward = (Baseline - Makespan) * Scale)
```

```
# ------------------
# 日志与监控 (Logging)
# ------------------
log_dir = "tf-logs"                    # TensorBoard 日志目录
```


Filter tags (regex)
All
Scalars
Image
Pinned
Pin cards for a quick view and comparison
Eval 2 cards
Eval/Balance_Std
Run
Value
Ste
ALB_PPO_20260219_201104 1,287.1519 96
Eval/Makespan
Run
Value
Ste
ALB_PPO_20260219_201104 3,217.8796 96
Loss 4 cards
Loss/Entropy
Run
Value
Step
ALB_PPO_20260219_201104 -0.167 96
Loss/Policy
Run
Value
Step
ALB_PPO_20260219_201104 0.0613 96
Loss/Total
Run
Value
Step
ALB_PPO_20260219_201104 4.9848 96
Loss/Value
Run
Value
Step
ALB_PPO_20260219_201104 5.2589 96
Reward
Reward/Episode
Run
Value
Step
ALB_PPO_20260219_201104 758.0177 100
Train
Train/LearningRate
Run
Value
Step
ALB_PPO_20260219_201104 0 96

4.7 以以后的版本已经可以算作失败了。打算退回 4.6，并进行修改。